1. Algebra and functions

What students need to learn:

Simplification of rational expressions including factorising and cancelling, and algebraic division.	Denominators of rational expressions will be linear or quadratic, e.g. $\dfrac{1}{ax+b}, \dfrac{ax+b}{px^2+qx+r}, \dfrac{x^3+1}{x^2-1}$.
Rational functions. Partial fractions (denominators not more complicated than repeated linear terms).	Partial fractions to include denominators such as $(ax+b)(cx+d)(ex+f)$ and $(ax+b)(cx+d)^2$. The degree of the numerator may equal or exceed the degree of the denominator. Applications to integration, differentiation and series expansions. Quadratic factors in the denominator such as (x^2+a), $a > 0$, are *not* required.
Definition of a function. Domain and range of functions. Composition of functions. Inverse functions and their graphs.	The concept of a function as a one-one or many-one mapping from $\mathbb{R}$ (or a subset of $\mathbb{R}$) to $\mathbb{R}$. The notation $\mathrm{f}: x \mapsto$ and $\mathrm{f}(x)$ will be used. Students should know that fg will mean 'do g first, then f'. Students should know that if f^{-1} exists, then $\mathrm{f}^{-1}\mathrm{f}(x) = \mathrm{f}\mathrm{f}^{-1}(x) = x$.
The modulus function.	Students should be able to sketch the graphs of $y = \lvert ax+b \rvert$ and the graphs of $y = \lvert \mathrm{f}(x) \rvert$ and $y = \mathrm{f}(\lvert x \rvert)$, given the graph of $y = \mathrm{f}(x)$.
Combinations of the transformations $y = \mathrm{f}(x)$ as represented by $y = a\mathrm{f}(x)$, $y = \mathrm{f}(x) + a$, $y = \mathrm{f}(x+a)$, $y = \mathrm{f}(ax)$.	Students should be able to sketch the graph of, for example, $y = 2\mathrm{f}(3x)$, $y = \mathrm{f}(-x) + 1$, given the graph of $y = \mathrm{f}(x)$ or the graph of, for example, $y = 3 + \sin 2x$, $y = -\cos\left(x + \dfrac{\pi}{4}\right)$. The graph of $y = \mathrm{f}(ax+b)$ will *not* be required.